

[Practical ML 이슈]

기계학습 리뷰

박성호

컴퓨터공학부

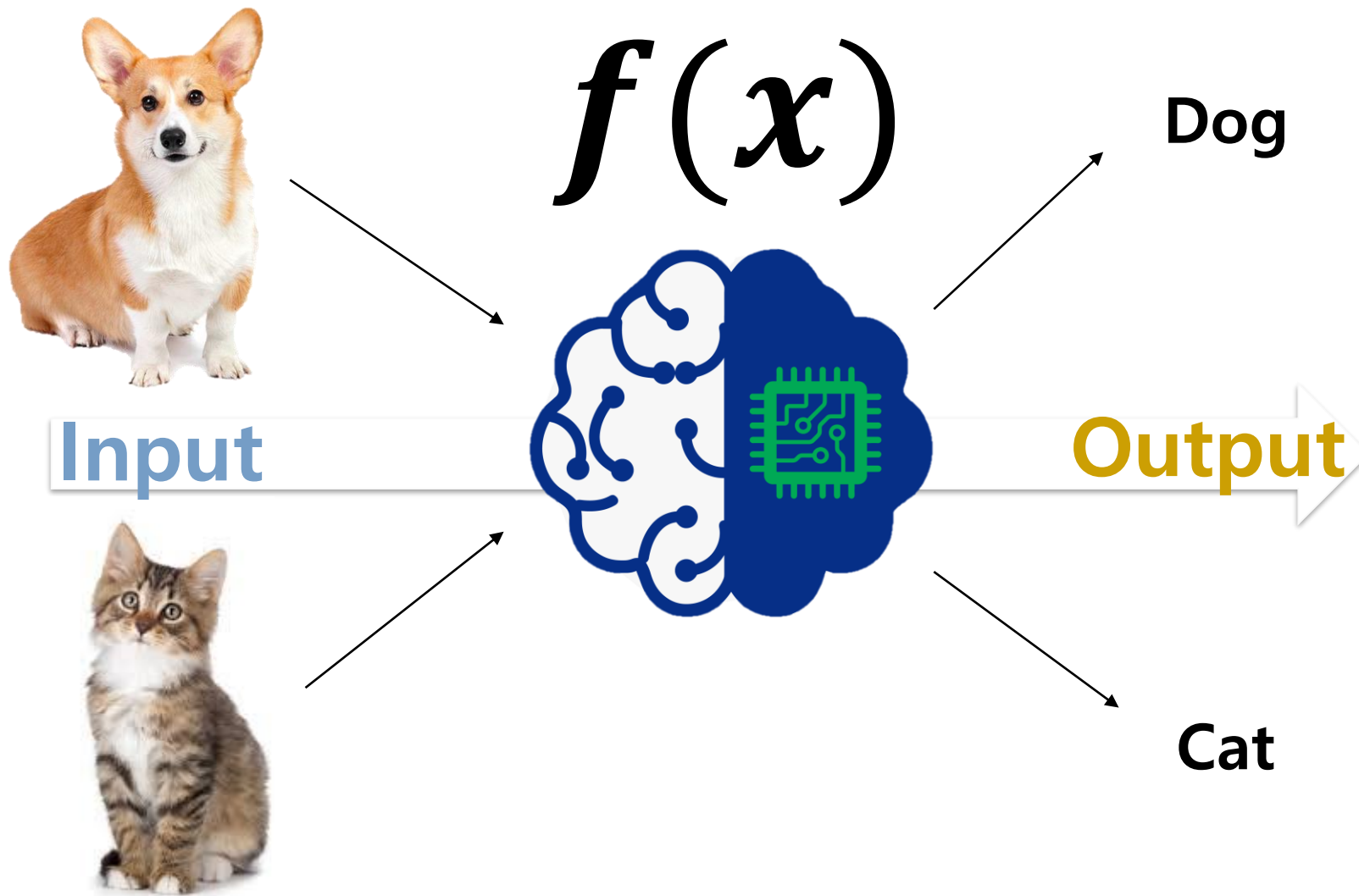
한국공학대학교

Agenda

- 1. 인공지능/기계학습이란?**
2. 인공지능에 따른 산업 변화
3. 기계학습 기본
 1. 회귀 문제를 위한 학습과정
 2. 분류 문제를 위한 학습과정
4. Toy example 시습

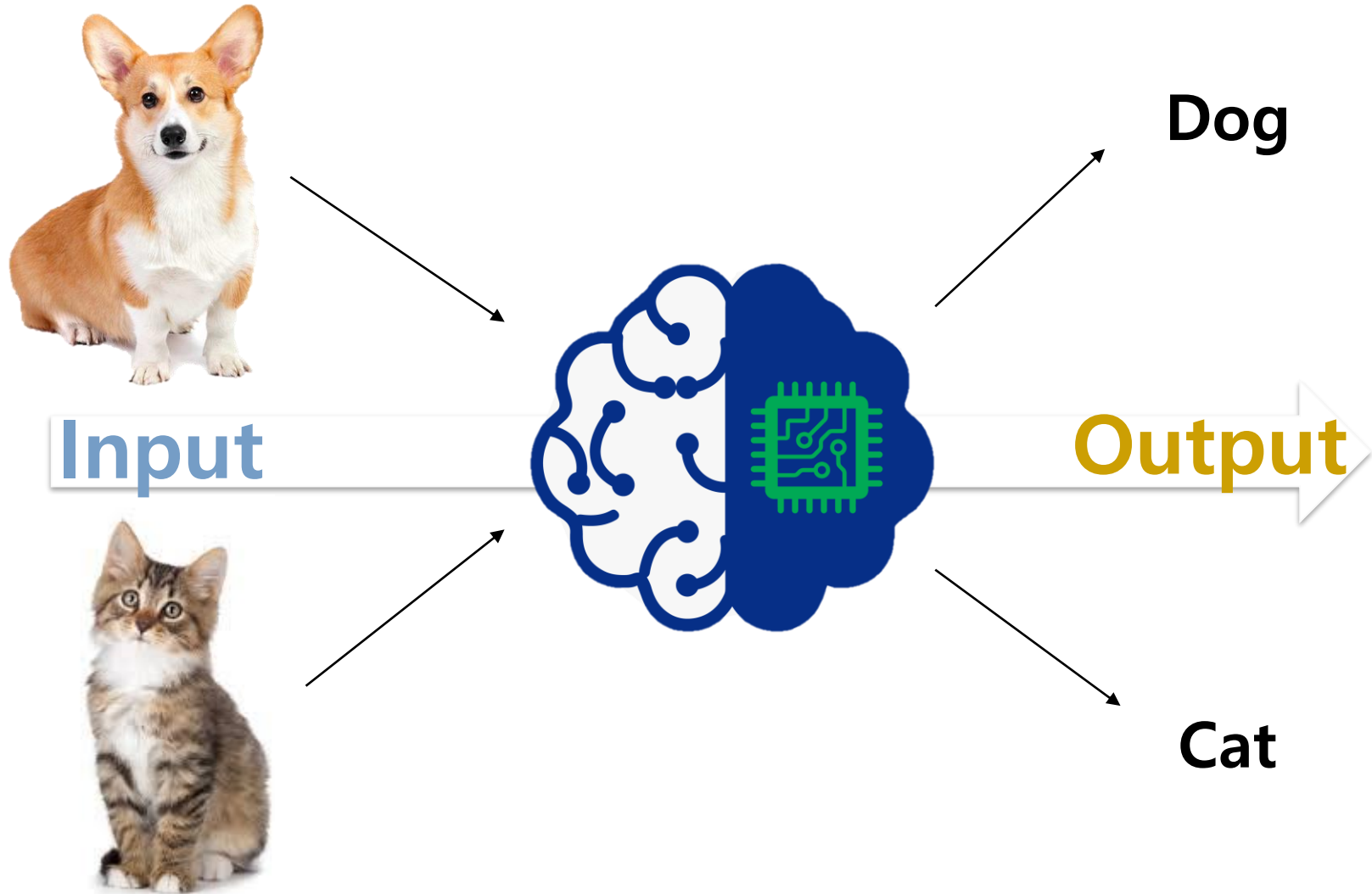
인공지능이란 무엇인가? – 쉬운 설명

❖ 인공지능의 핵심은 함수다?!



인공지능이란 무엇인가? – 쉬운 설명

❖ 인공지능은 입력값을 넣어주면 출력값이 나오는 시스템



인공지능이란 무엇인가? – 쉬운 설명

❖ 인공지능은 입력값을 넣어주면 출력값이 나오는 시스템

Input

NAVER 지도 검색

자동차 대중교통 자전거 도보

안암오거리교차로

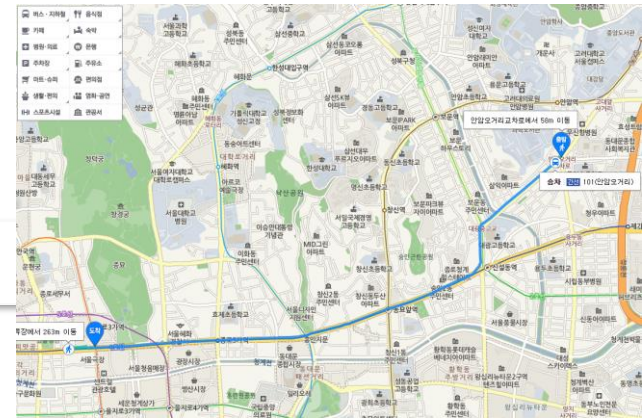
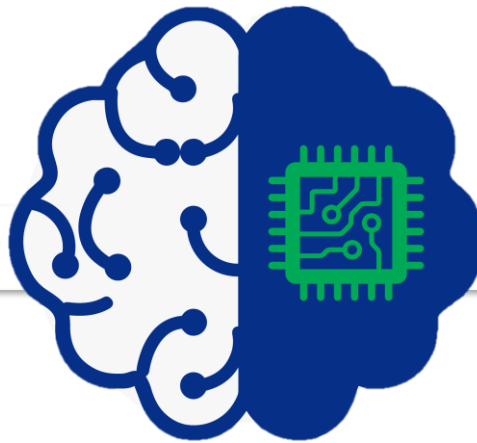
종로3가역 1호선

+ 경유지 추가 지우기

장소 버스

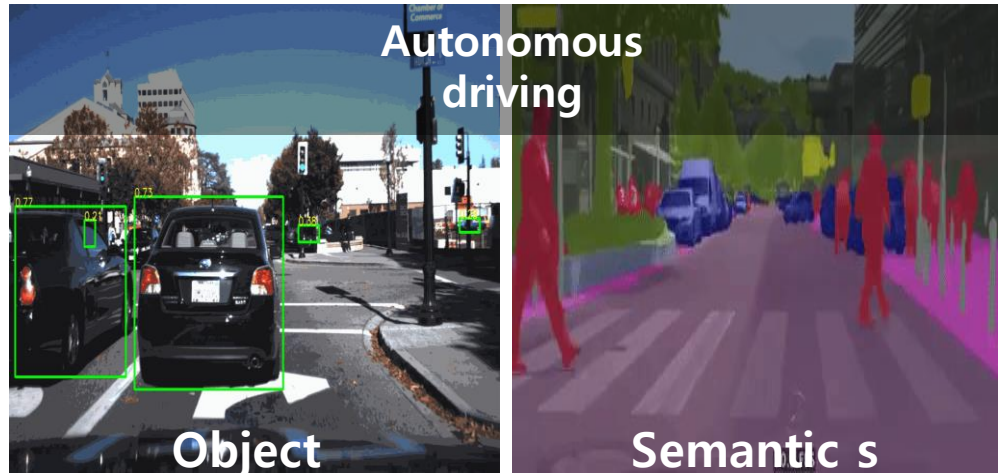
'종로3가역' 근처 장소 검색결과 66,018건

서울특별시 성북구 중심으로 '종로3가역' 검색하기

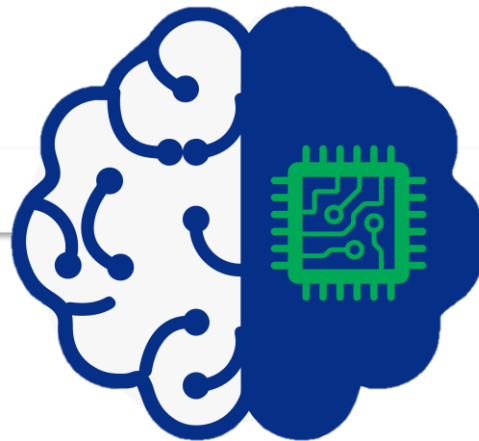


인공지능이란 무엇인가? – 쉬운 설명

❖ 인공지능은 입력값을 넣어주면 출력값이 나오는 시스템



Input



Output

인공지능이란 무엇인가? – 쉬운 설명

❖ 인공지능은 입력값을 넣어주면 출력값이 나오는 시스템

The screenshot shows a chat window titled "Default (GPT-3.5)". On the left is a sidebar with a "New chat" button and a list of "Previous 30 Days" chats. The main chat area shows a user asking a question about a research paper on self-attention and t-SNE. The AI responds with a detailed explanation of the concepts and suggests a literature search.

Previous 30 Days

- Variational Encoder in Informal
- July
- VAE Alignment Interpretation
- Stoch Backprop & Reparam Tr
- Attentive Reduction in CNN
- VAEs & Graphical Models.
- Gradient Length and Informati
- May
- Simplex constraint in optimiza
- April
- Self-attention and t-SN  
- Affine transformation explaine
- Unsupervised CAM method.
- Scaling in Linear Regression

Chat History:

User: let me know that is there the research paper about self-attention method using the idea of t-stochastic neighbor embedding

AI: I am not aware of any research paper that specifically combines self-attention methods with t-stochastic neighbor embedding (t-SNE). However, there are some papers that explore the use of both self-attention and t-SNE separately.

For instance, "Attention is All You Need" by Vaswani et al. (2017) introduced the Transformer model, which utilizes self-attention to perform tasks such as machine translation and language modeling.

On the other hand, t-SNE is a dimensionality reduction technique that is often used for visualizing high-dimensional data in lower dimensions. There are many papers that apply t-SNE to various domains, such as "Visualizing Data using t-SNE" by Maaten and Hinton (2008).

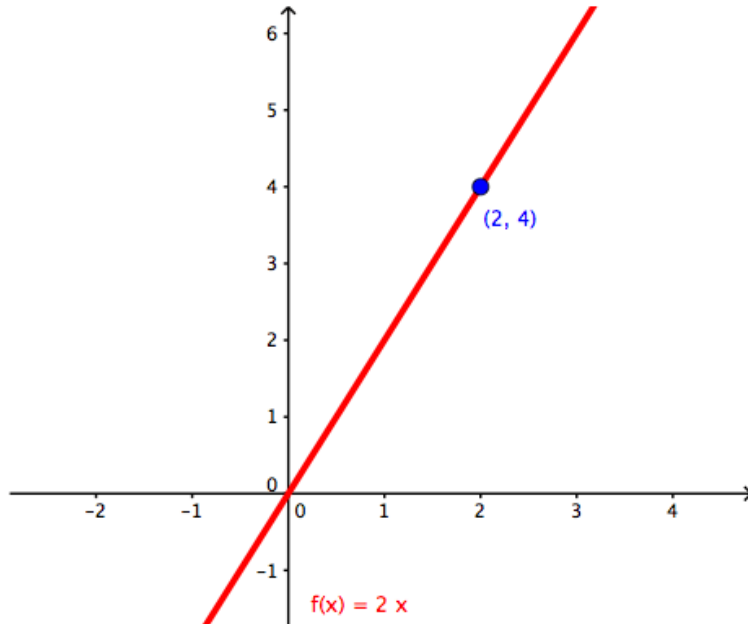
While there may not be a paper that directly combines self-attention and t-SNE, it is possible that some researchers have used both techniques in their work. I would suggest conducting a literature search on the topic of interest to see if any relevant papers exist.

Bottom Bar:

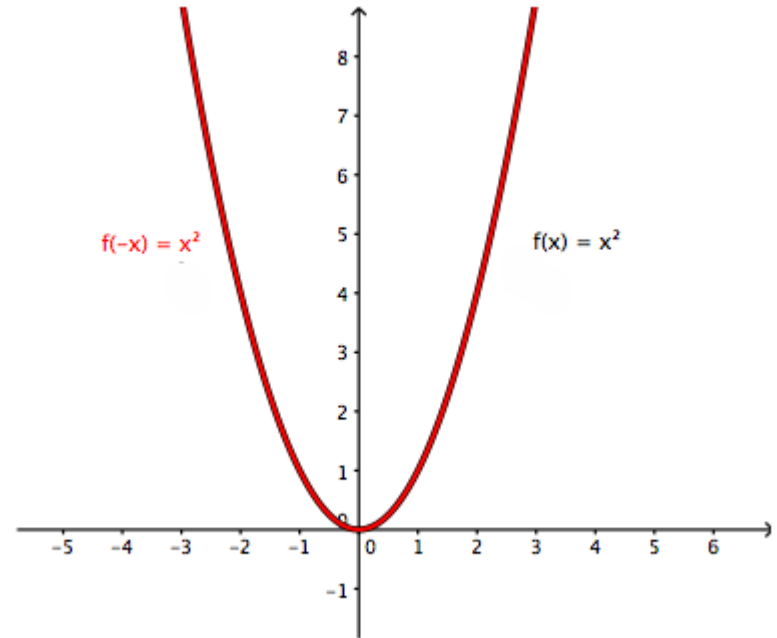
- Lightshot Screenshot
- Send a message
- Regenerate
- Upgrade to Plus
- NEW

인공지능이 함수라고?!

- 파이썬으로 함수를 짜봅시다.



```
def linear_function(x):  
    input = x  
    output = 2*x  
    return output
```



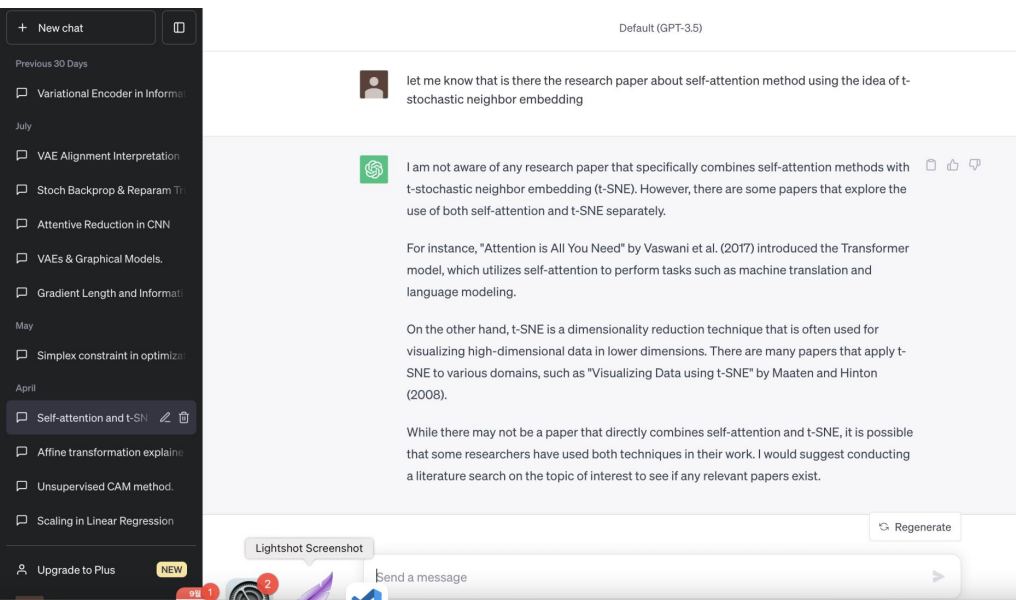
```
def quadratic_function(x):  
    input = x  
    output = x**2  
    return output
```

기계학습 (Machine Learning)?

❖ 우리 주변의 상황들을 모델링할 수 있는 모든 함수를 알 수 있을까?

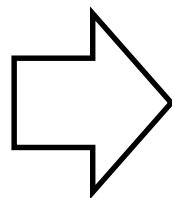
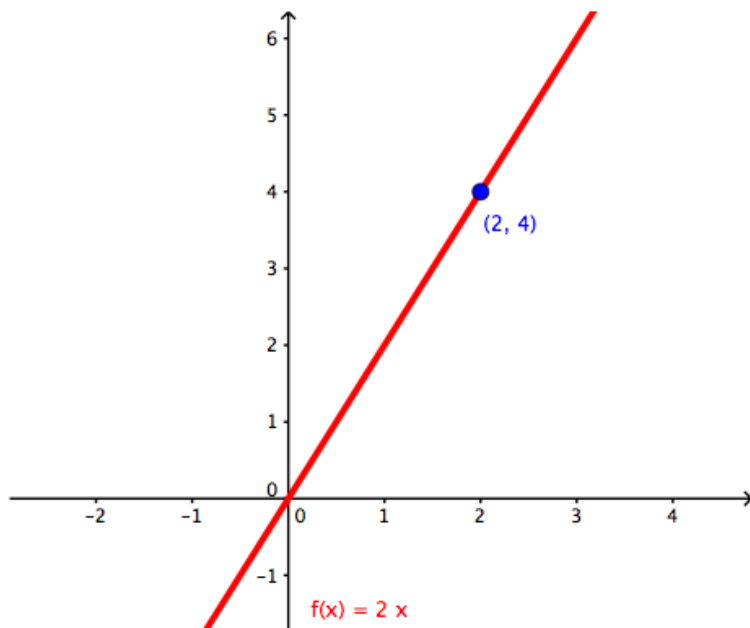
$$f(x)$$

질문에 대해
논리적으로
그리고 사실에
근거해서
때로는
상상력을
더해서?



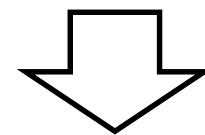
기계학습 (Machine Learning)?

- ❖ 문제를 보는 새로운 관점이 필요 – 사람이 함수를 알 수 없다면, 데이터를 이용해서 컴퓨터가 스스로 함수를 찾게 하자!



데이터	
1	2
2	4
3	6
4	8
5	10
6	12

$$ax = y$$

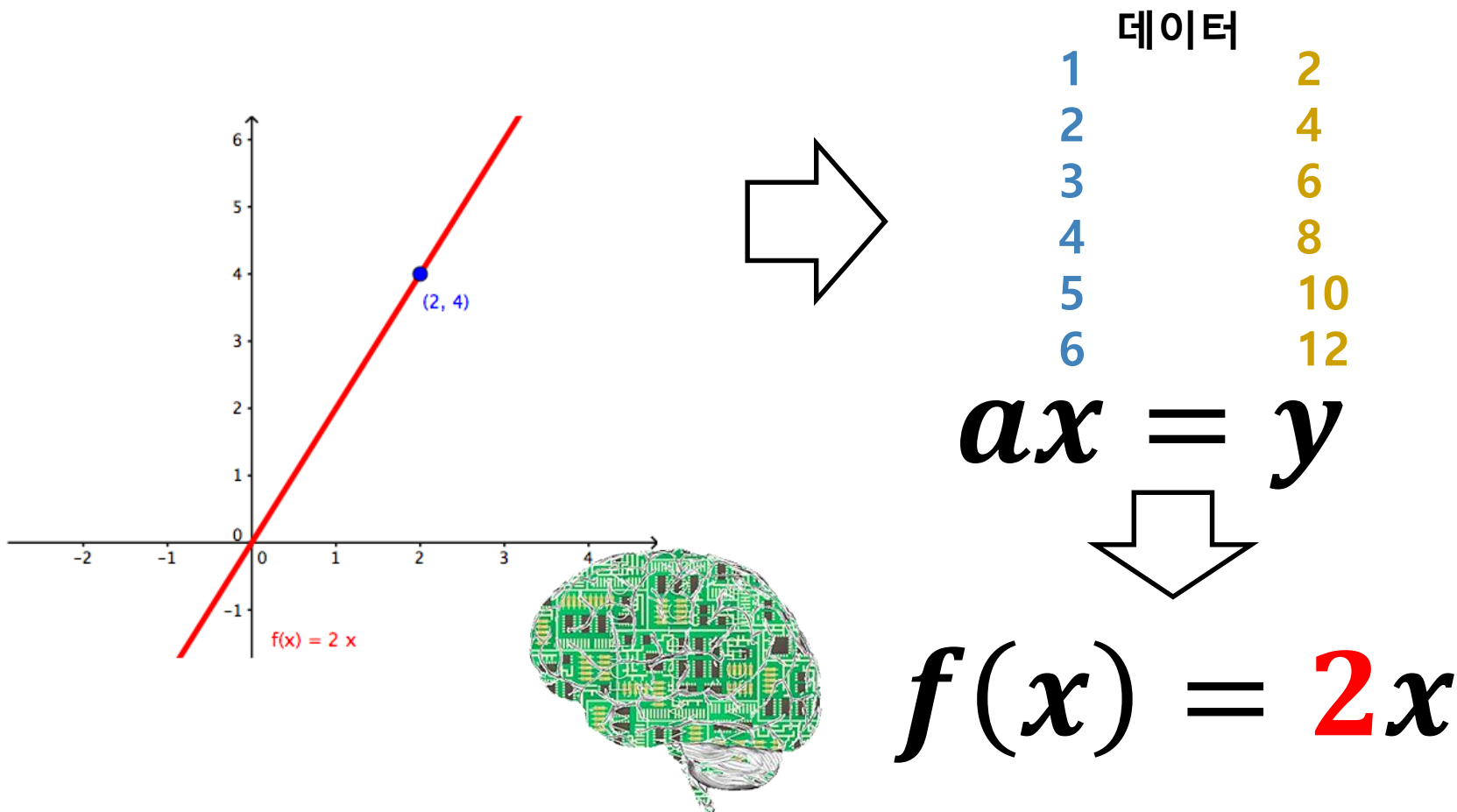


$$f(x) = 2x$$



기계학습 (Machine Learning)?

- ❖ 인공지능, 머신러닝에 대한 잘못된 환상 – 데이터만 있으면 모델이 알아서 생성되고 학습된다?! – No – 기본이 되는 모델/알고리즘 형태는 사람이 결정, 입력



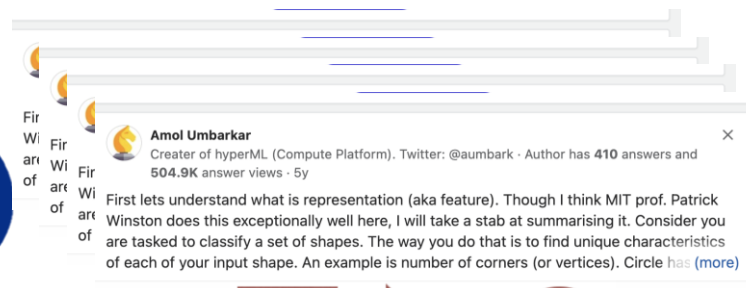
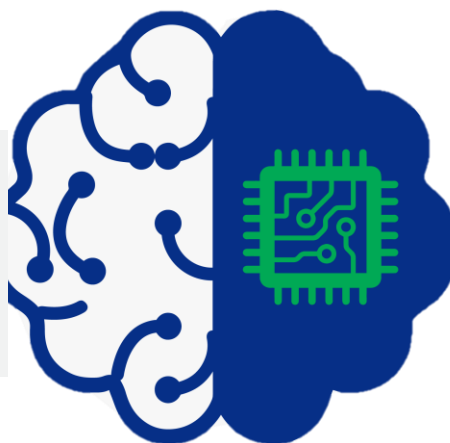
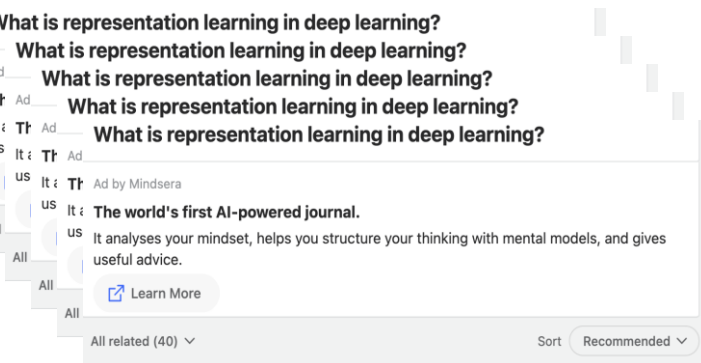
기계학습 (Machine Learning)?

- ❖ 문제를 보는 새로운 관점이 필요 – 사람이 함수를 알 수 없다면, 데이터를 이용해서 컴퓨터가 스스로 함수를 찾게 하자!

Input

Output

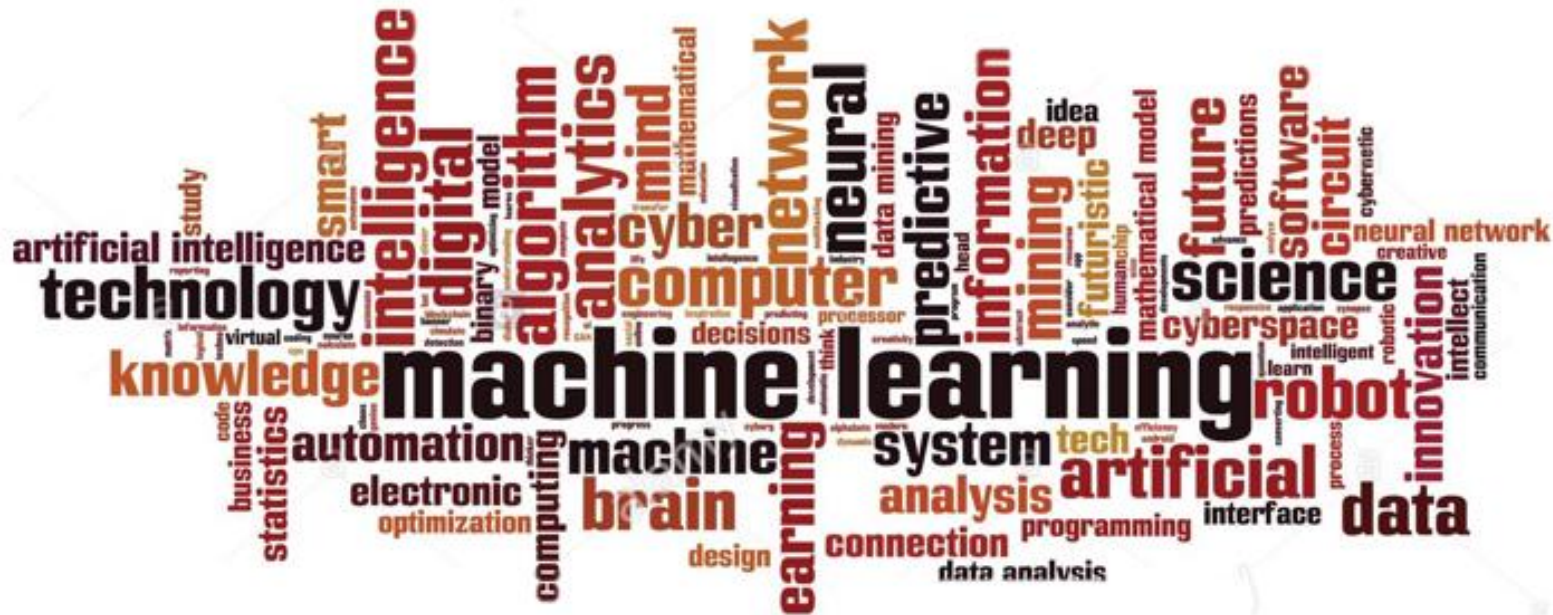
Learning!



$f_{chatgtp}(x)$ with deep learning

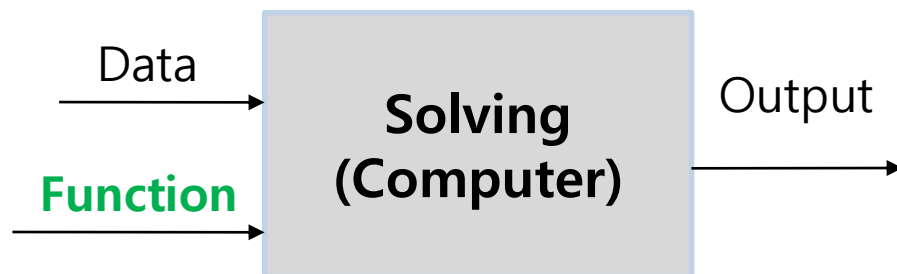
기계학습 (Machine Learning)?

- Definition: an application of AI that provides systems the ability to automatically learn and improve from experience without being explicitly programmed



기계학습 (Machine Learning)?

- 일반적인 프로그래밍 방식과의 차이점
 - 일반적인 컴퓨터 프로그램



- 머신러닝 프로그램

(3) Algorithm

(2) Data

**(1) Concept
(Computer)**

Training

Function

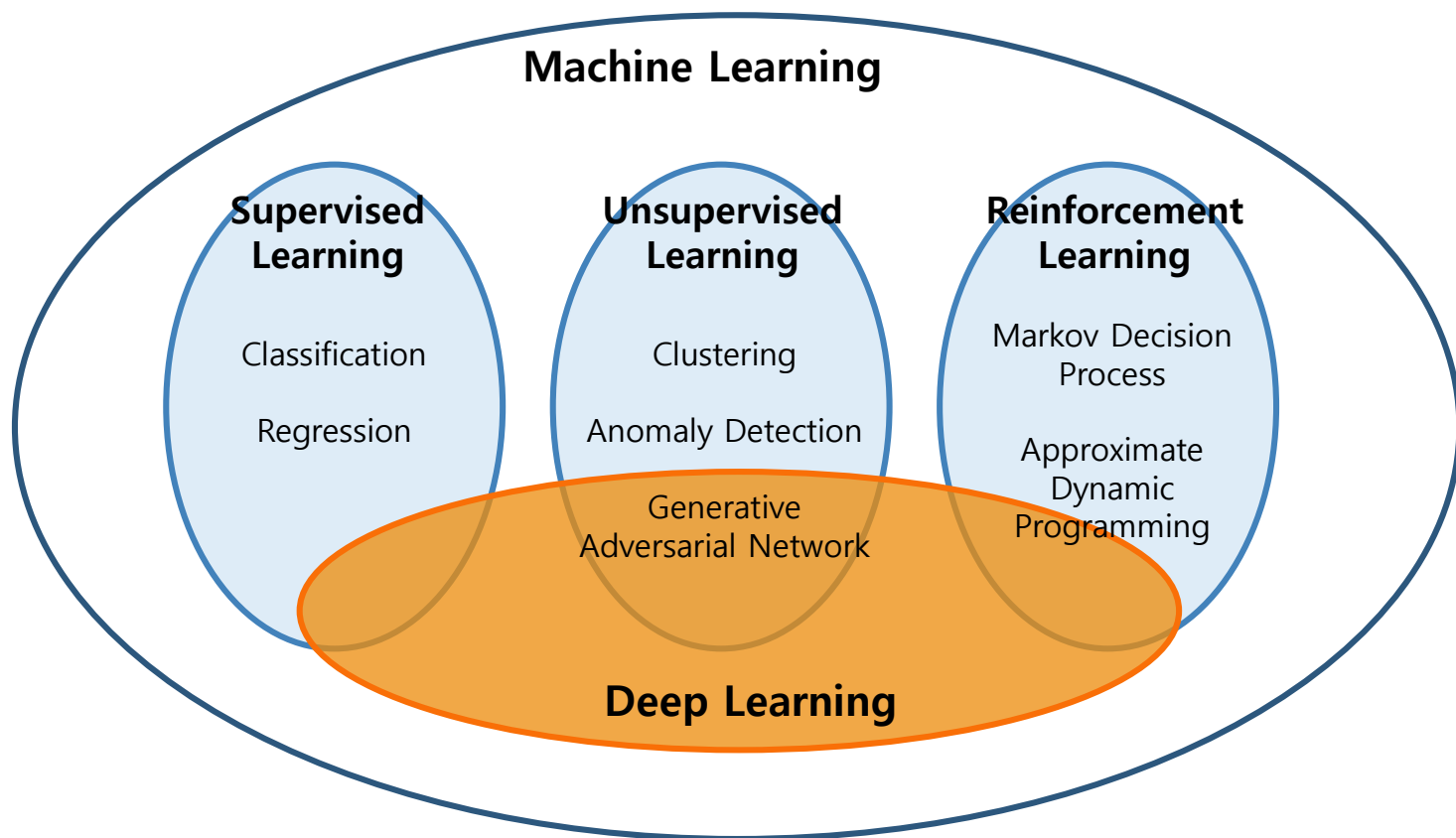
**Solving
(Computer)**

Testing

Output

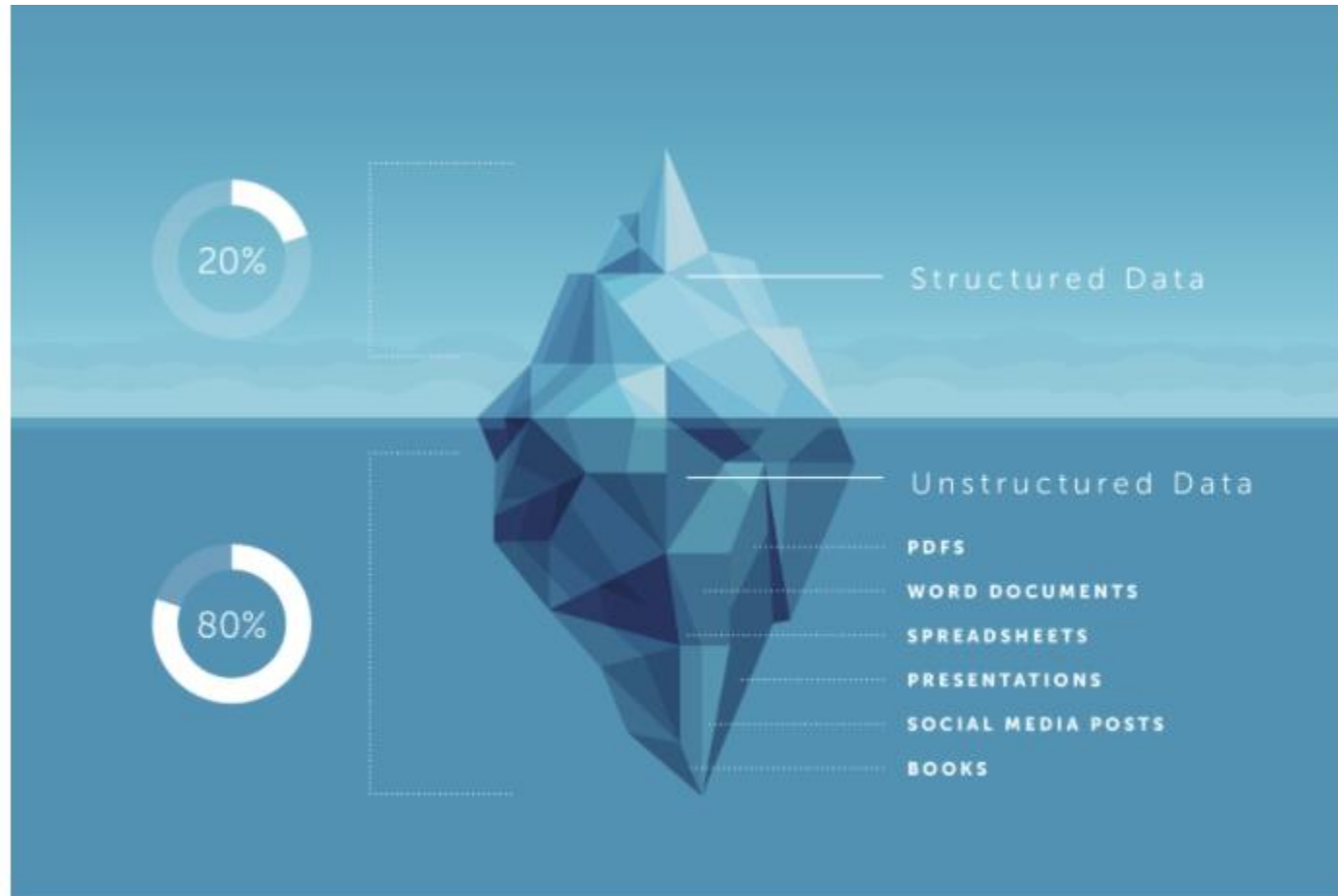
기계학습 범주

- Deep learning은 machine learning에 속하는 방법론 중 하나
- 본 강의에서는 supervised learning의 관점에서 deep learning을 다룸



What Kind of Data ?

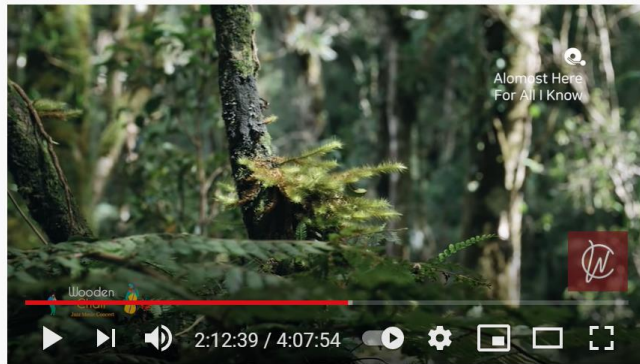
❖ Structured and Unstructured



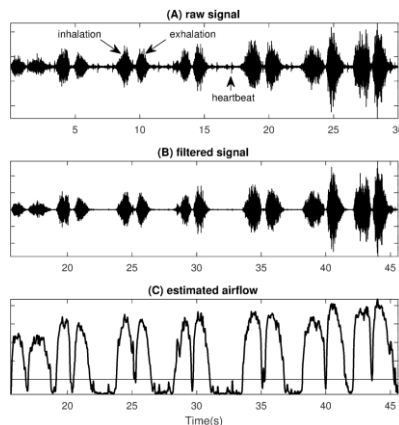
What Kind of Data ?

- ❖ Structured and **Unstructured**
- ❖ **cannot** be processed and analyzed using conventional tools and method
- ❖ 데이터의 의미를 쉽게 파악 x

Video



Signal (음파, 진동)



Text

Mode-wise Tensor Decompositions: Multi-dimensional Generalizations of CUR Decompositions

HanQin Cai

Department of Mathematics
University of California, Los Angeles
Los Angeles, CA 90095, USA

HQCAI@MATH.UCLA.EDU

Keaton Hamm

Department of Mathematics
University of Texas at Arlington
Arlington, TX 76019, USA

KEATON.HAMM@UTA.EDU

Longxiu Huang

Department of Mathematics
University of California, Los Angeles
Los Angeles, CA 90095, USA

HUANGL3@MATH.UCLA.EDU

Deanna Needell

Department of Mathematics
University of California, Los Angeles
Los Angeles, CA 90095, USA

DEANNA@MATH.UCLA.EDU

Editor: Michael Mahoney

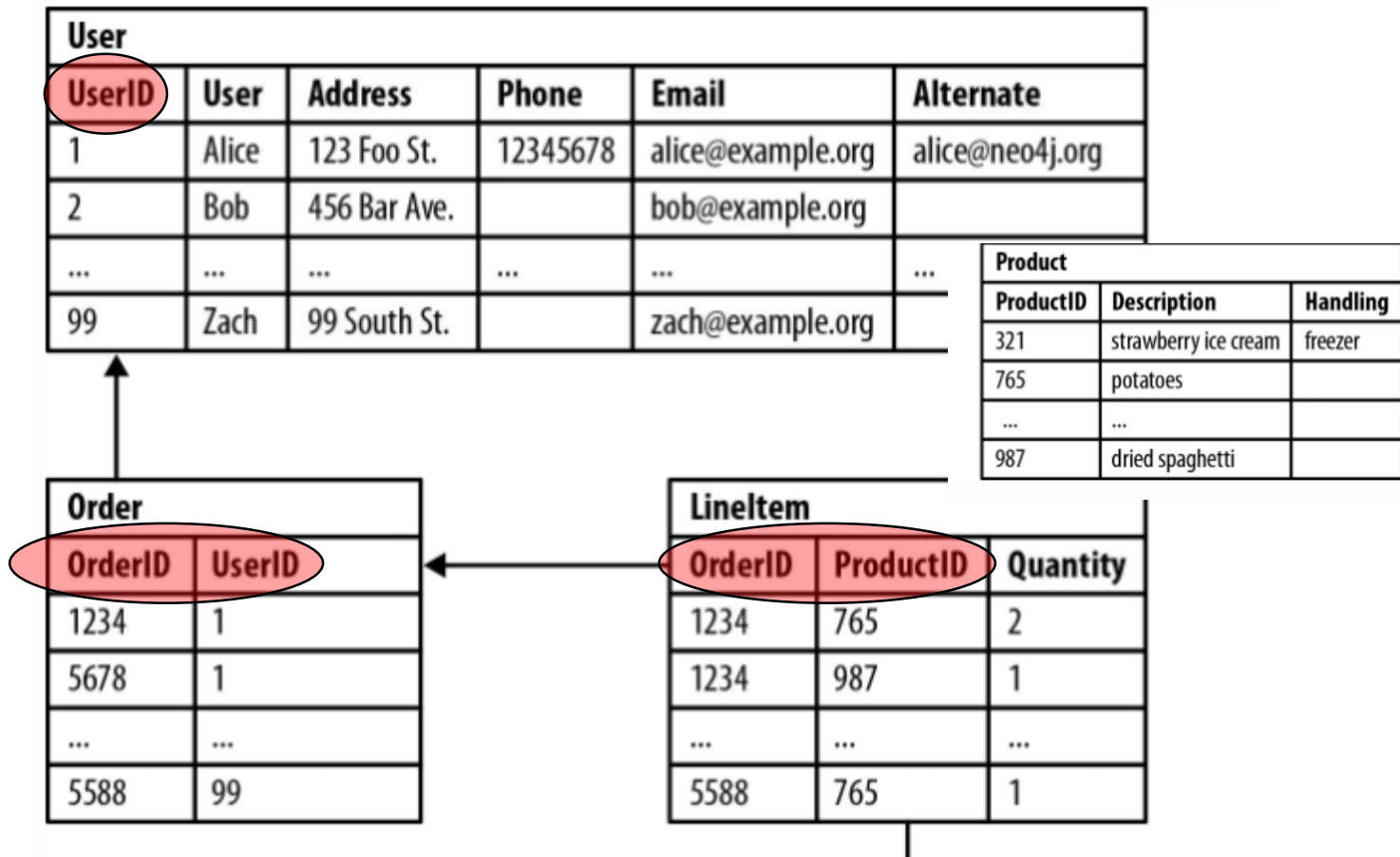
Abstract

Low rank tensor approximation is a fundamental tool in modern machine learning and data science. In this paper, we study the characterization, perturbation analysis, and an efficient sampling strategy for two primary tensor CUR approximations, namely Chidori and Fiber CUR. We characterize exact tensor CUR decompositions for low multilinear rank tensors. We also present theoretical error bounds of the tensor CUR approximations when (adversarial or Gaussian) noise appears. Moreover, we show that low cost uniform sampling is sufficient for tensor CUR approximations if the tensor has an incoherent structure. Empirical performance evaluations, with both synthetic and real-world datasets, establish the speed advantage of the tensor CUR approximations over other state-of-the-art low multilinear rank tensor approximations.

Keywords: tensor decomposition, low-rank tensor approximation, CUR decomposition, randomized linear algebra, hyperspectral image compression

What Kind of Data ? (전처리)

- ❖ Structured and Unstructured
- ❖ Structured data is highly organized
- ❖ relational databases and spreadsheets (키와 값으로 구성된 테이블)
- ❖ 컬럼을 통해 데이터의 의미를 쉽게 파악



기계학습에 사용되는 데이터

정형 데이터

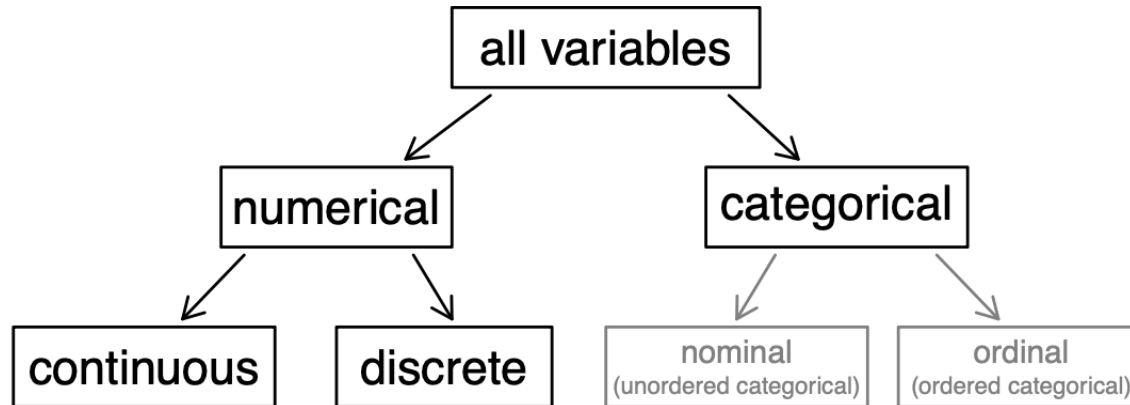
- 테이블과 같이 고정된 컬럼에 저장되는 데이터와 파일, 그리고 행과 열에 의해 데이터의 속성이 구별되는 스프레드시트 형태의 데이터

변수: 각 관측치의 특성치 (Variable, Feature, Factor)

관측치: sample (Observation, Sample, Pattern)

	Sepal.Length	Sepal.Width	Petal.Length	Petal.Width	Species
1	5.1	3.5	1.4	0.2	setosa
2	4.9	3.0	1.4	0.2	setosa
3	4.7	3.2	1.3	0.2	setosa
4	4.6	3.1	1.5	0.2	setosa
5	5.0	3.6	1.4	0.2	setosa
6	5.4	3.9	1.7	0.4	setosa
7	4.6	3.4	1.4	0.3	setosa
8	5.0	3.4	1.5	0.2	setosa
9	4.4	2.9	1.4	0.2	setosa
10	4.9	3.1	1.5	0.1	setosa

기계학습에 사용되는 데이터



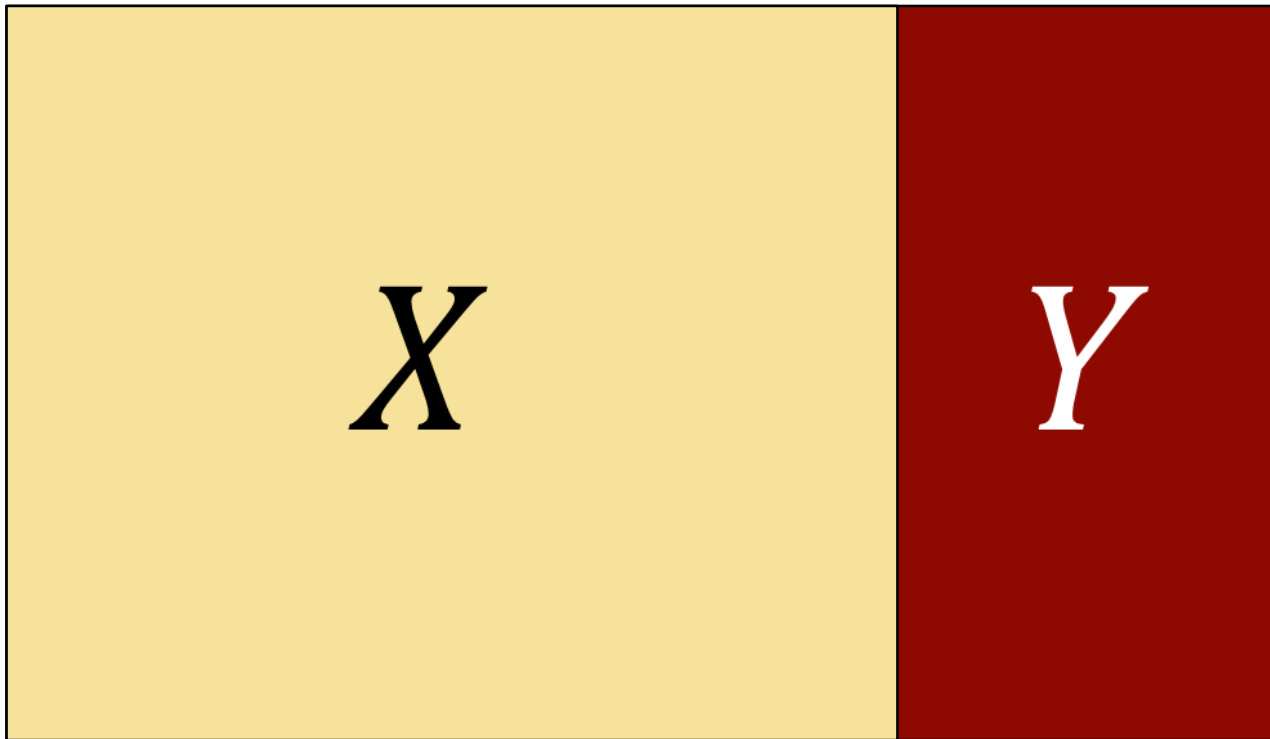
■ 범주형 변수 (Categorical Variable)

- 명목형 (Nominal): 여러 Class들 중 하나의 이름에 데이터를 분류할 수 있을 때 사용 – 예를 들어 사과, 배, 바나나 구분
- 순서형 (Ordinal): 데이터가 속하는 Class들에 순서가 있는 경우 순서형 (ordinal) 데이터라고 한다. – 예를 들어 상, 중, 하 구분

■ 수치형 변수 (Numerical Variable)

- 연속형 (Continuous): 실수 형태의 연속적인 값을 가지고 있는 변수를 연속형 변수라 한다. – 예를 들어 키나 몸무게와 같은 변수
- 이산형 (Discrete): 정수 형태의 값을 갖고 있는 변수 – 예를 들어 투표수, 인구수 등....

기계학습에 사용되는 데이터



독립변수 (Independent variable)

예측변수 (Explanatory variable)

인풋변수 (Input variable)

종속변수 (Dependent variable)

반응변수 (Response variable)

아웃풋변수 (Output variable)

기계학습에 사용되는 데이터

중고차 데이터

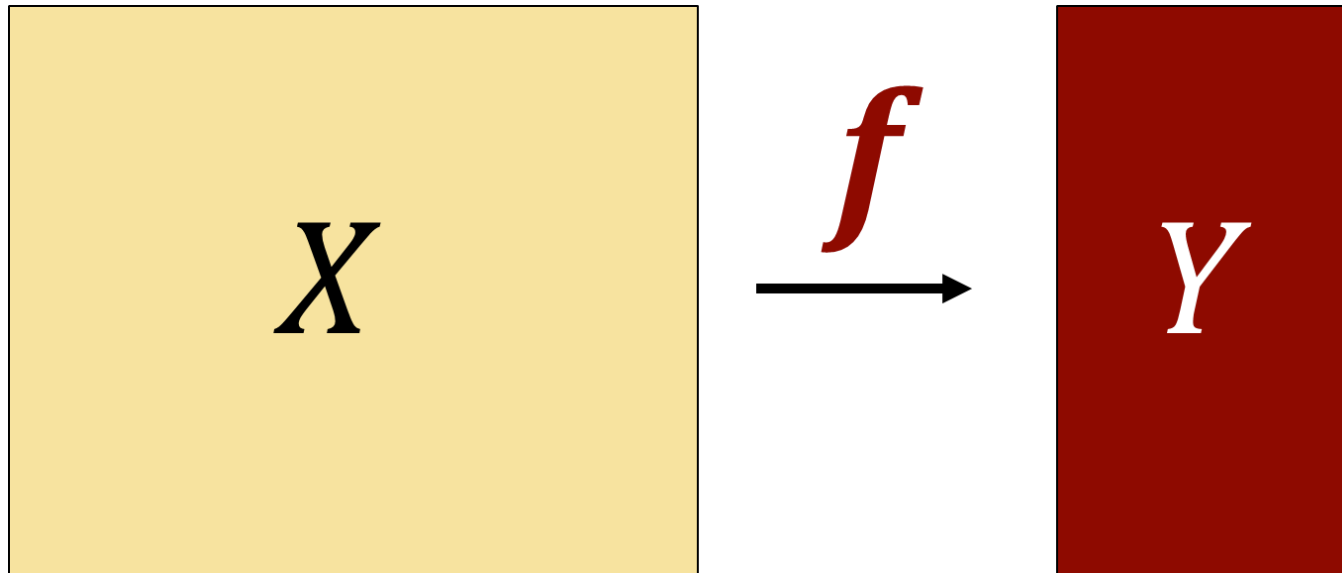
X Y

모델	주행거리	마력	용량	가격
TOYOTA Corolla 2.0 D4D HATCHB TERRA 2/3-Doors	46,986	90	2,000	13,500
TOYOTA Corolla 1800 T SPORT VVT I 2/3-Doors	19,700	192	1,800	21,500
TOYOTA Corolla 1.9 D HATCHB TERRA 2/3-Doors	71,138	69	1,900	12,950
TOYOTA Corolla 1.8 VVTL-i T-Sport 3-Drs 2/3-Doors	31,461	192	1,800	20,950
TOYOTA Corolla 1.8 16V VVTLI 3DR T SPORT BNS 2/3-Doors	43,610	192	1,800	19,950
TOYOTA Corolla 1.6 VVTI Linea Terra Comfort 2/3-Doors	21,716	110	1,600	17,950
TOYOTA Corolla 1.6 16v L.SOL 2/3-Doors	25,563	110	1,600	16,750
TOYOTA Corolla 1.6 16V VVT I 3DR TERRA 2/3-Doors	64,359	110	1,600	16,950
TOYOTA Corolla 1.6 16V VVT I 3DR SOL AUT4 2/3-Doors	43,905	110	1,600	16,950
TOYOTA Corolla 1.6 16V VVT I 3DR SOL 2/3-Doors	56,349	110	1,600	15,950
TOYOTA Corolla 1.4 VVTI Linea Terra 2/3-Doors	9,750	97	1,400	12,950
TOYOTA Corolla 1.4 16V VVT I 3DR 2/3-Doors	27,500	97	1,400	14,750
TOYOTA Corolla 1.4 16V VVT I 3DR 2/3-Doors	49,059	97	1,400	13,950
TOYOTA Corolla 1.4 16V VVT I 3DR 2/3-Doors	44,068	97	1,400	16,750
TOYOTA Corolla 1.4 16V VVT I 3DR 2/3-Doors	46,961	97	1,400	13,950
TOYOTA Corolla 2.0 D4D 90 5DR TERRA COMFORT 4/5-Doors	110,404	90	2,000	16,950
TOYOTA Corolla 2.0 D4D 90 5DR TERRA COMFORT 4/5-Doors	100,250	90	2,000	16,950
TOYOTA Corolla 2.0 D4D 90 5DR SOL 4/5-Doors	84,000	90	2,000	19,000
TOYOTA Corolla 2.0 D4D 90 5DR TERRA 4/5-Doors	79,375	90	2,000	17,950
TOYOTA Corolla 1.4 16V VVT I 5DR TERRA COMFORT 4/5-Doors	75,048	97	1,400	15,800

지도 학습 (Supervised Learning)

- 학습데이터로부터 $X \rightarrow Y$ 의 관계를 설명하는 함수 (f)를 찾는 방법론
- 종속변수 y 가 범주형이면 분류(classification), 연속형이면 회귀(regression)

$$f(x) \approx y$$



지도 학습의 두 가지 모델 방법론 (분류)

- CIFAR10 dataset



지도 학습의 두 가지 모델 방법론 (회귀)

중고차 데이터

$X \quad f(x) \quad Y$

모델	주행거리	마력	용량	가격
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지도학습 (Supervised Learning)

핵심 아이디어...

우리의 주 관심은 Y (예측하려는 대상)

Y 를 설명하는 X 변수는 보통 여러 개

여러 개의 X 와 Y 의 관계를 찾는 것!

X 변수들을 조합(결합)하여 Y 를 표현

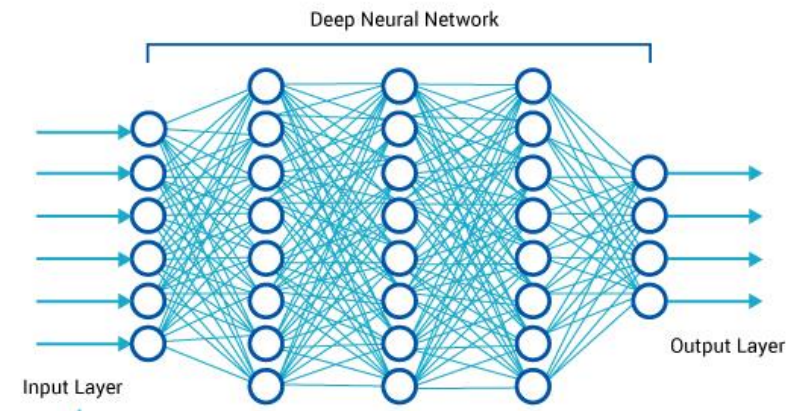
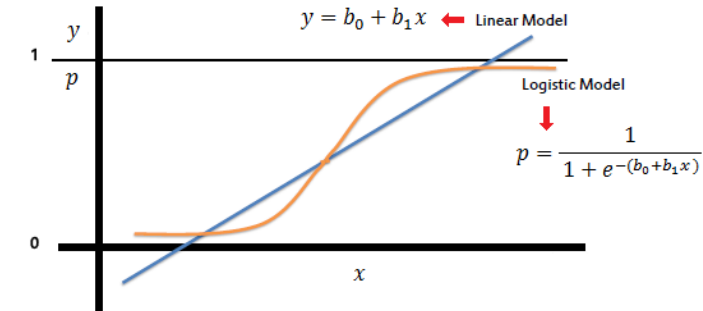
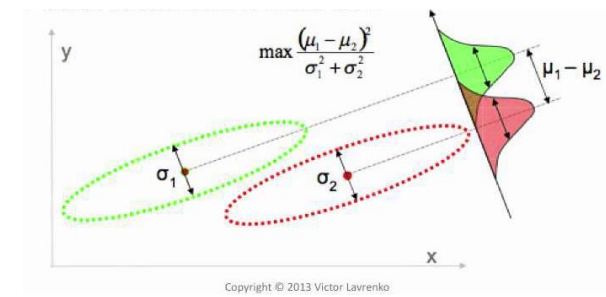
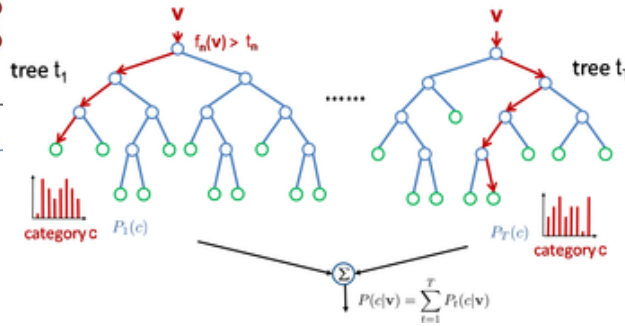
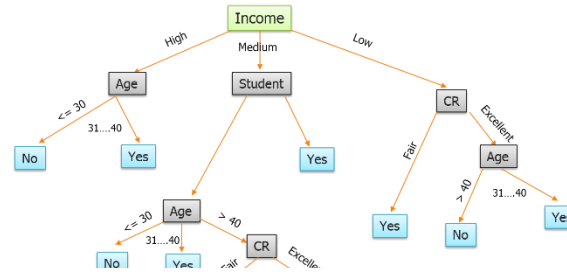
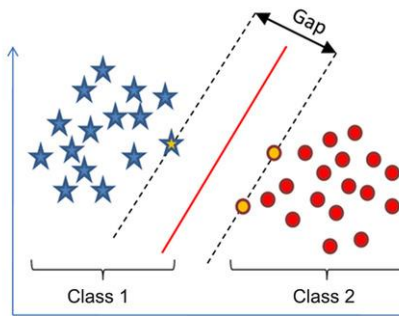
조합하는 방법은 무수히 많음

수학적으로는, $Y = f(X_1, X_2, \dots, X_p)$

f

Supervised Learning Models

- 함수 F 의 형태에 따라 다양한 모델이 존재

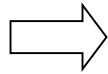


의사결정나무 모델

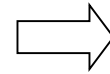
- 데이터에 내재되어 있는 패턴을 변수의 조합으로 나타내는 예측/분류 모델을 나무의 형태로 만드는 것
- 질문을 던져서 맞고 틀리는 것에 따라 우리가 생각하고 있는 대상을 좁혀나감
- “스무고개”놀이와 비슷한 개념

의사결정나무 모델 개요

Data



Algorithm

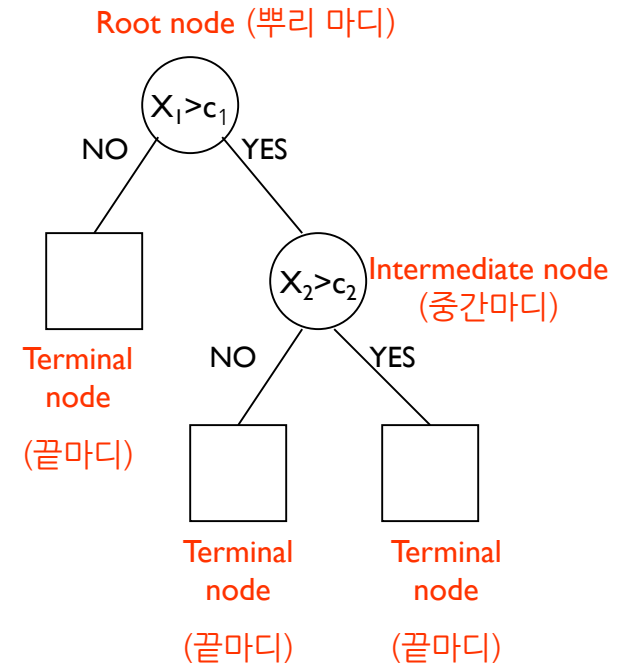


Model (Output)

Input				Output
X_1	X_2	...	X_p	Y

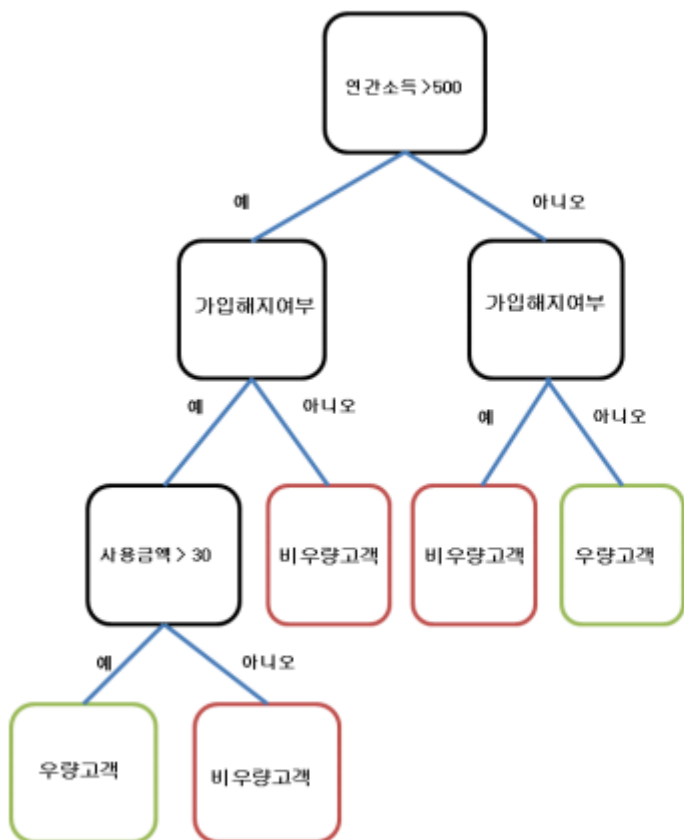
- 데이터를 2개 혹은 그 이상의 부분집합으로 분할
- 분류: 비슷한 범주를 갖고 있는 관측치끼리 모음
- 예측: 비슷한 수치를 갖고 있는 관측치끼리 모음

- Parent node: 분할 전의 노드
- Child node: 분할 후의 노드
- Root node: 처음 시작 노드 - child node만 가지고 있음
- Leaf nodes: 마지막 노드 - parent node만 가지고 있음
- Split criterion: 노드를 분할하는데 사용한 변수값
예) 가입해지여부 = 예

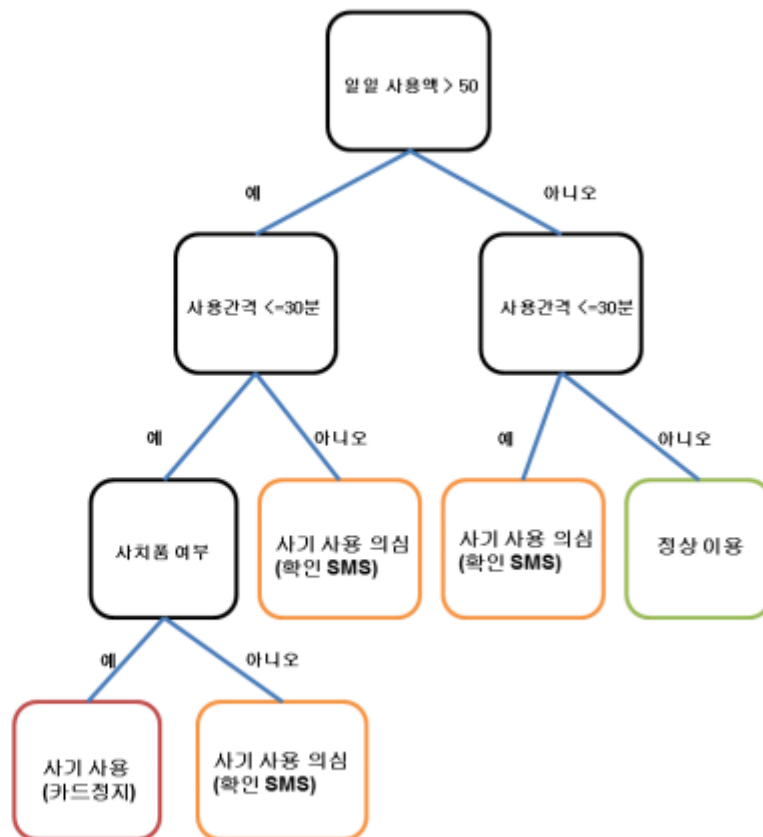


의사결정나무 모델 예시

1) 우량고객 평가 모델 (Customer value evaluation)



2) 카드 사기 방지 모델 (Fraud detection)



왜 사용할까?

- 간단하며 직관적이고 결과에 대한 해석이 가능하다
- 분류 의사결정나무(Classification Tree)는 범주형 반응변수를 예측하는데 사용
- 회귀 의사결정나무(Regression Tree)는 연속형 반응변수를 예측하는데 사용

Classification Tree

What we make learner to learn ?

예제)

문제: "쉬는 시간 X_1 " 과 "공부시간 X_2 "에 따른 학원 성취도 (Good=초록/Bad=빨강)

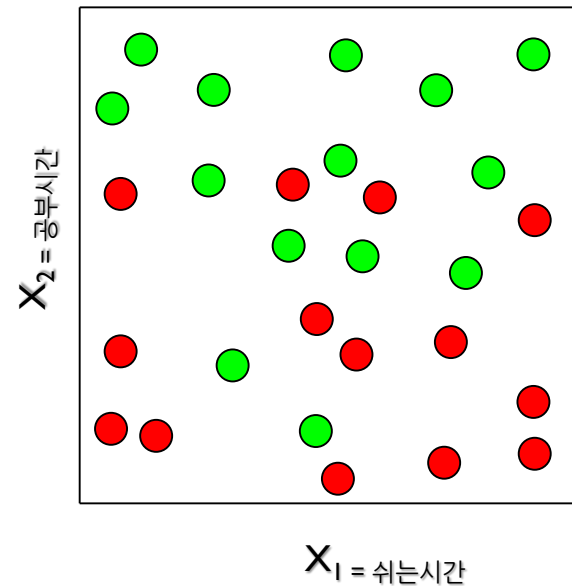
Input Variables: "쉬는 시간 X_1 " 과 "공부시간 X_2 "

Input Types: Continuous Variables

Output Variable: 학업 성취도 (good/bad)

Output Type: Discrete Variable-이진

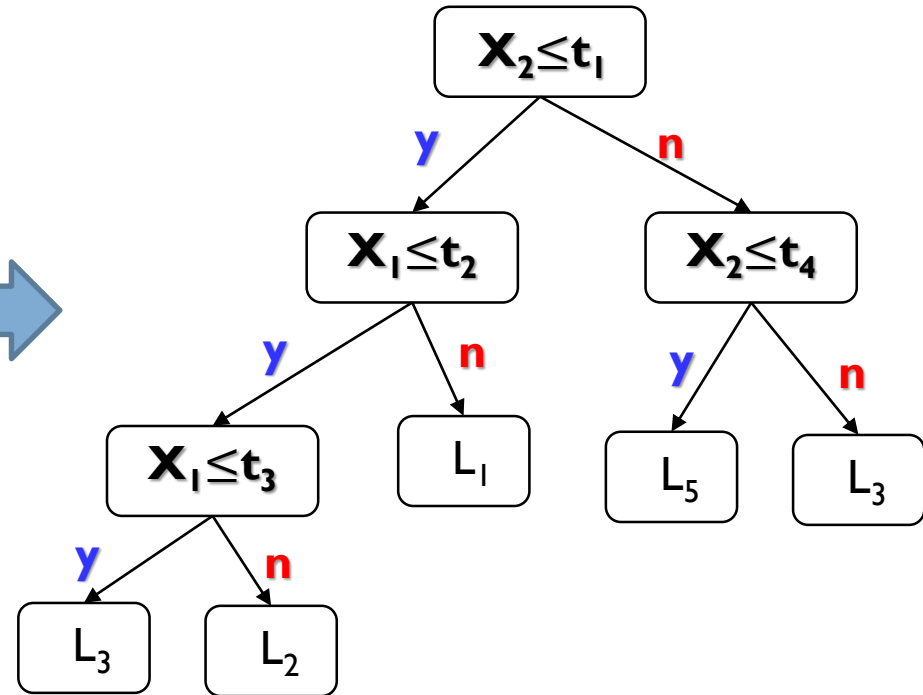
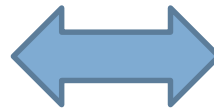
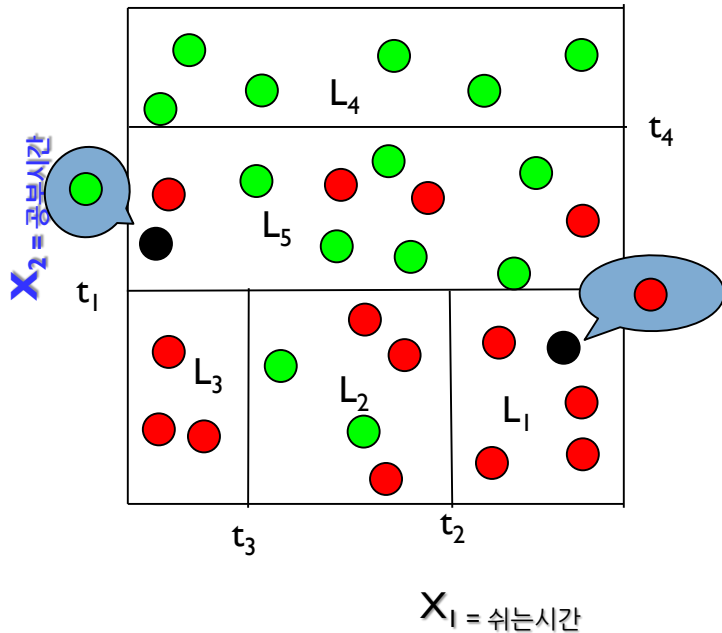
- Criterion = 어떤 인풋, 어떤 값, 얼마큼?
- How can we find or decide these?
- How can it function for prediction?



Classification Tree – Split input space (making if-then rule)

입력 변수의 영역을 두 개로 구분

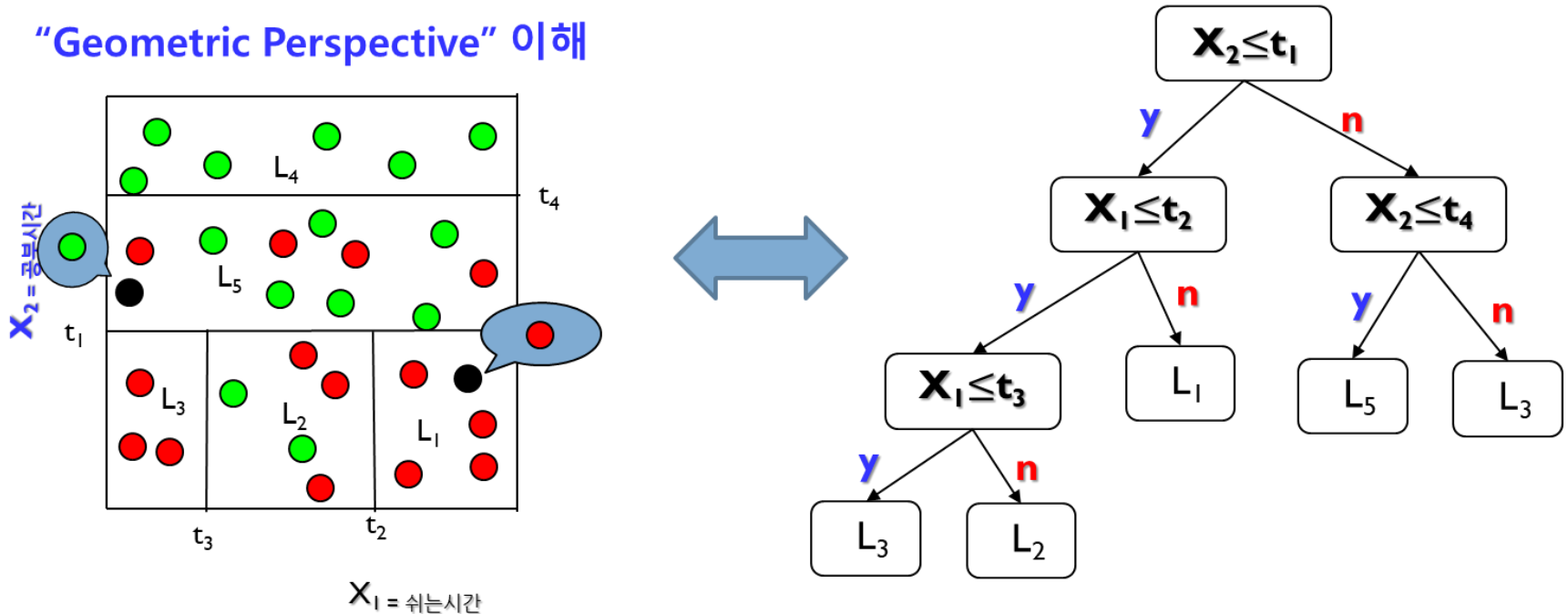
“Geometric Perspective” 이해



Classification Tree – Split input space (making if-then rule)

입력 변수의 영역을 두 개로 구분

“Geometric Perspective” 이해



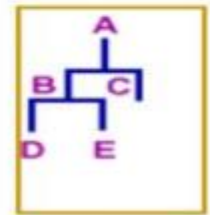
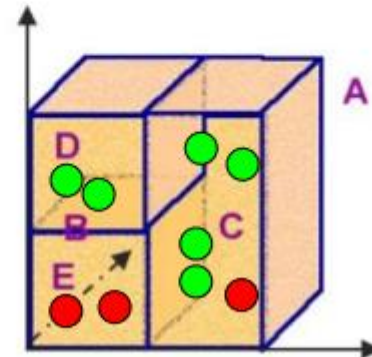
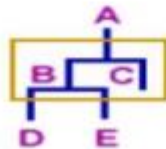
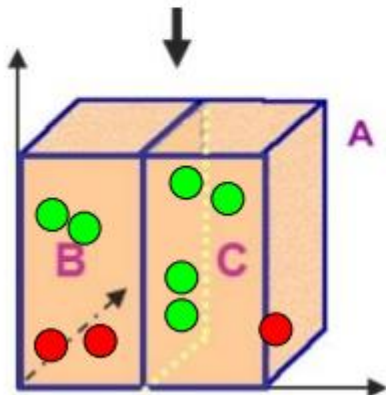
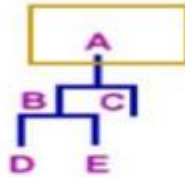
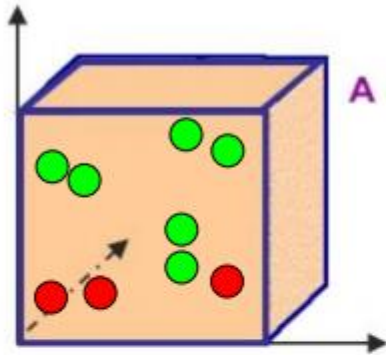
Good=1 $\approx p(y=1) = 6/10 = f(x_{\text{test1}}) = f(x_1:10, x_2:50)$

Bad=0 $\approx p(y=1) = 0/4 = f(x_{\text{test2}}) = f(x_1:90, x_2:30)$

데이터가 속한 공간에서 특정 클래스가 나올 확률 계산
Classification 진행

Classification Tree – How to split space (making if-then rule)

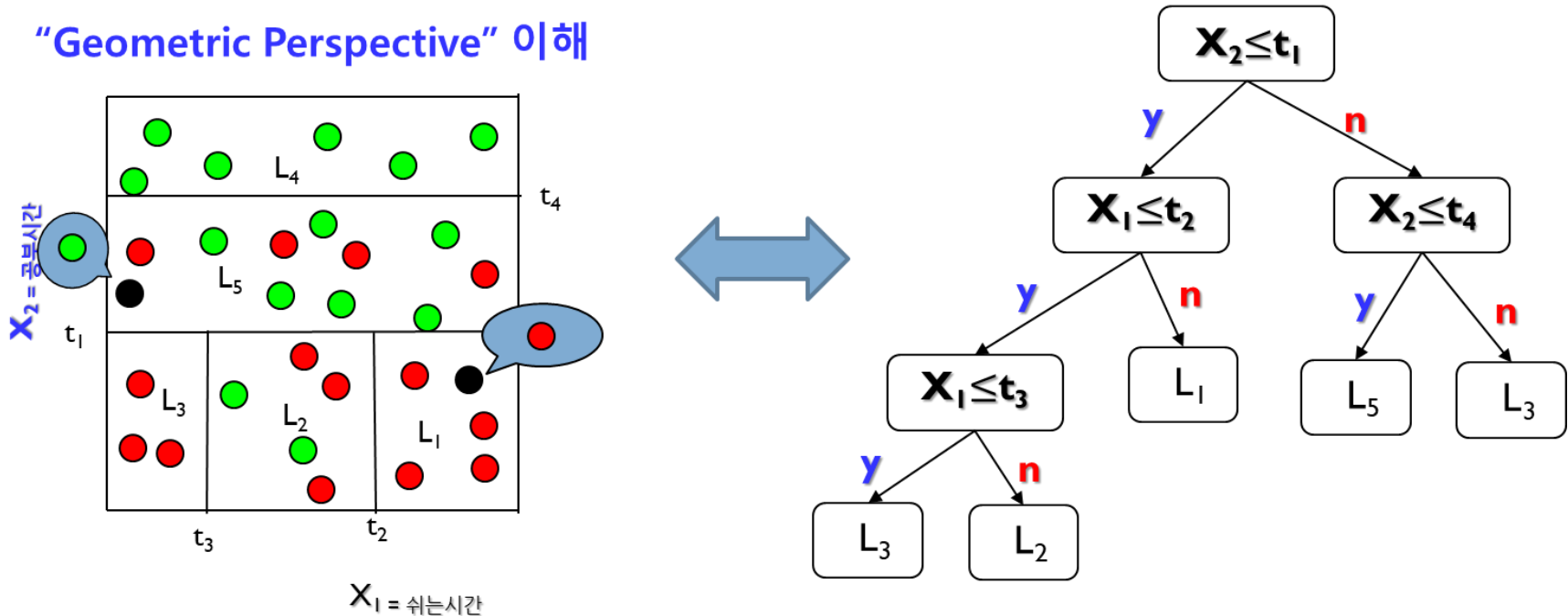
입력 변수의 영역을 두 개로 구분



Classification Tree – Split input space (making if-then rule)

입력 변수의 영역을 두 개로 구분

“Geometric Perspective” 이해



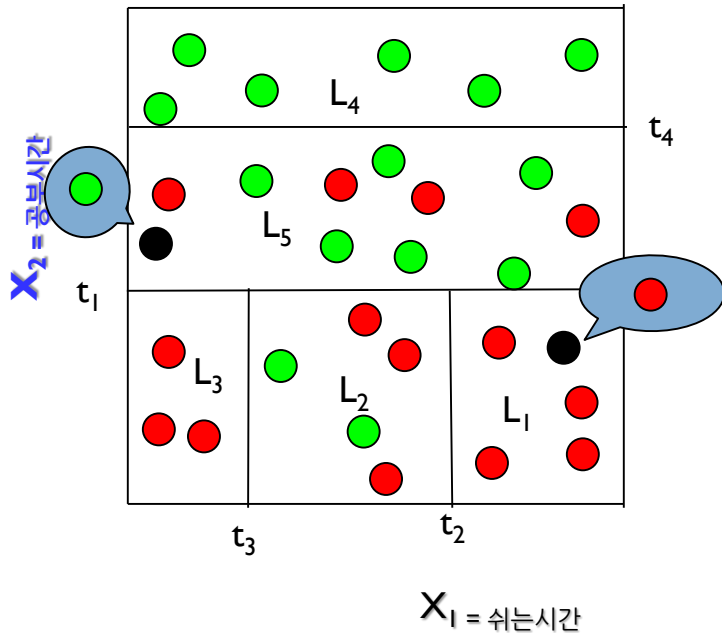
Good=1 $\approx p(y=1) = 6/10 = f(x_{\text{test1}}) = f(x_1:10, x_2:50)$

Bad=0 $\approx p(y=1) = 0/4 = f(x_{\text{test2}}) = f(x_1:90, x_2:30)$

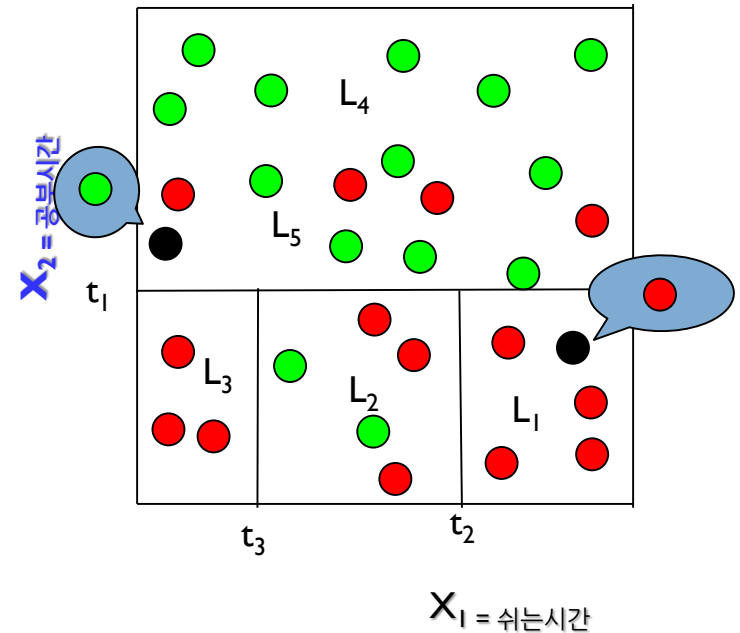
데이터가 속한 공간에서 특정 클래스가 나올 확률 계산
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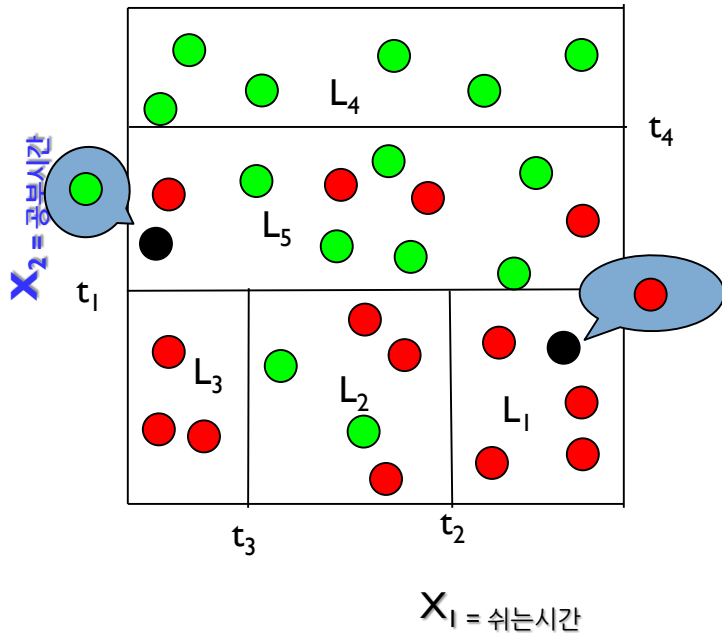


$f(x_{\text{test1}}) = f(x_1:10, x_2:50)$
 $f(x_{\text{test2}}) = f(x_1:90, x_2:30)$

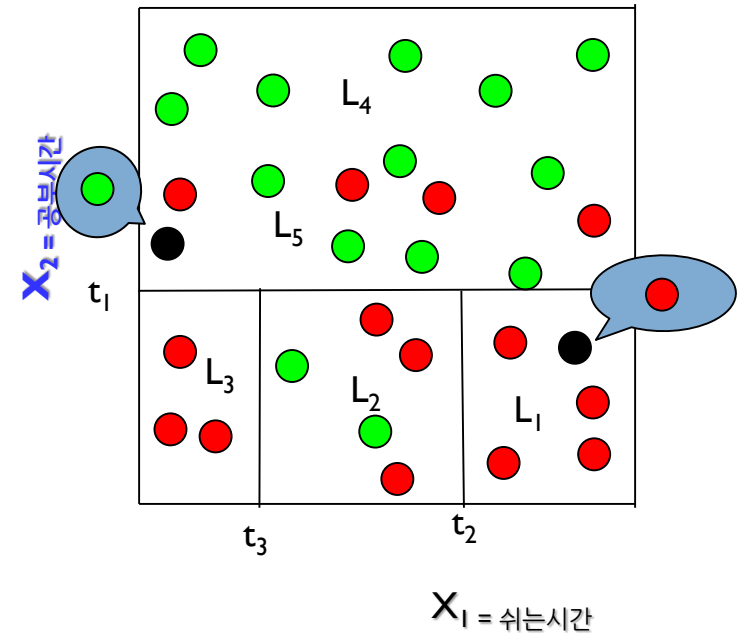
데이터가 속한 공간에서 특정 클래스가 나올 확률 계산
Classification 진행

Classification Tree – Split input space (making if-then rule)

입력 변수의 영역을 두 개로 구분



Good=1 $\approx p(y=1) = 6/10 = f(x_{\text{test1}}) = f(x1:10, x2:50)$
 Bad=0 $\approx p(y=1) = 0/4 = f(x_{\text{test2}}) = f(x1:90, x2:30)$

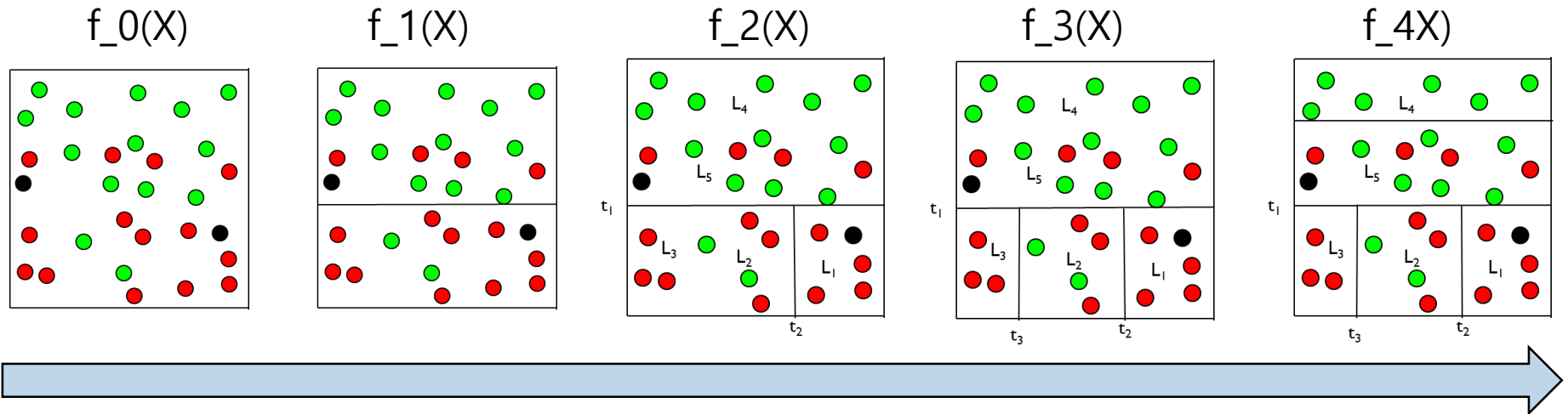


Good=1 $\approx p(y=1) = 13/17 = f(x_{\text{test1}}) = f(x1:10, x2:50)$
 Bad=0 $\approx p(y=1) = 0/4 = f(x_{\text{test2}}) = f(x1:90, x2:30)$

데이터가 속한 공간에서 특정 클래스가 나올 확률 계산
 Classification 진행

Classification Tree – Split input space (making if-then rule)

입력 변수의 영역을 두 개로 구분

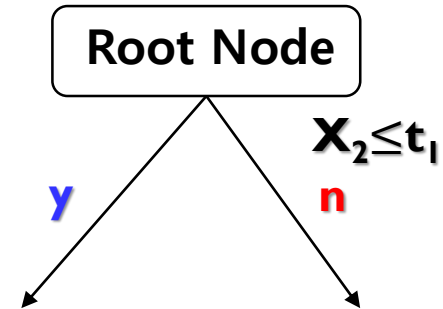
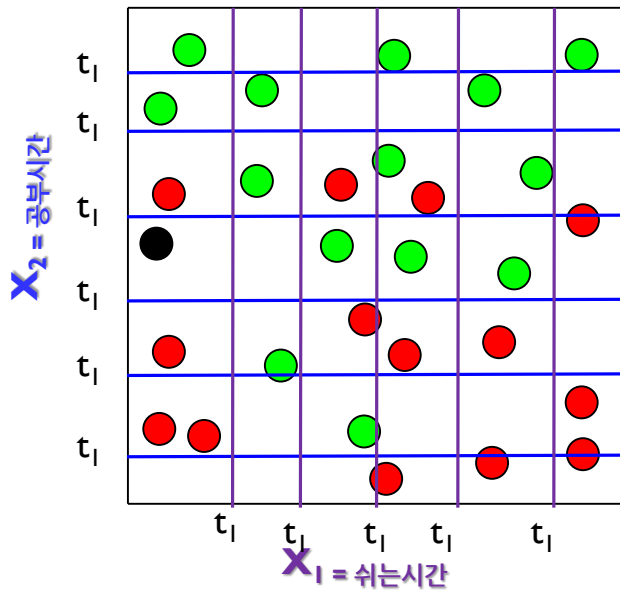


점진적으로 f 가 업그레이드

Classification을 더 잘 그리고 더 확실하게, 신뢰도 높게 판단

Classification Tree – How to split space (making if-then rule)

“Geometric Perspective”



- Why we choose this variable and value for partitioning?
- Just search all candidates? Why not?
- Greedy search

We need “Split Criterion



Classification Tree - Cost/Loss 비용/손실 (Impurity Measure)Function

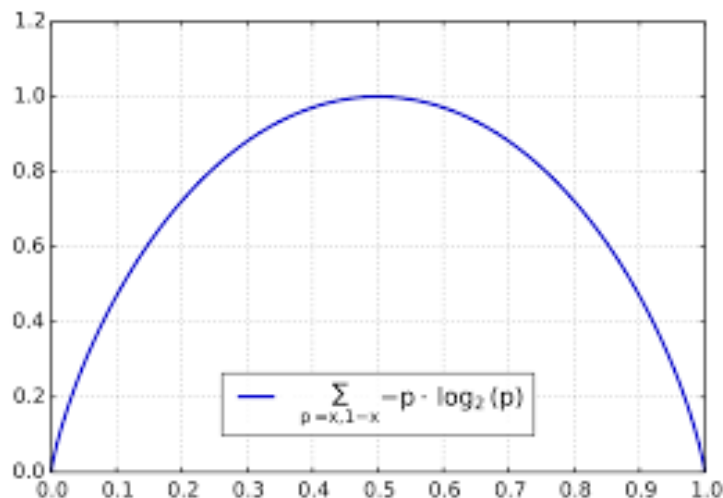
- 분류 모델에서의 비용함수 (불순도 측정)

In **information theory**, the **entropy** of a **random variable** is the average level of "information", "surprise", or "uncertainty" inherent to the variable's possible outcomes. Given a discrete random variable X , which takes values in the alphabet $\mathcal{X}$ and is distributed according to $p : \mathcal{X} \rightarrow [0, 1]$:

$$H(X) := - \sum_{x \in \mathcal{X}} p(x) \log p(x)$$

Ex:

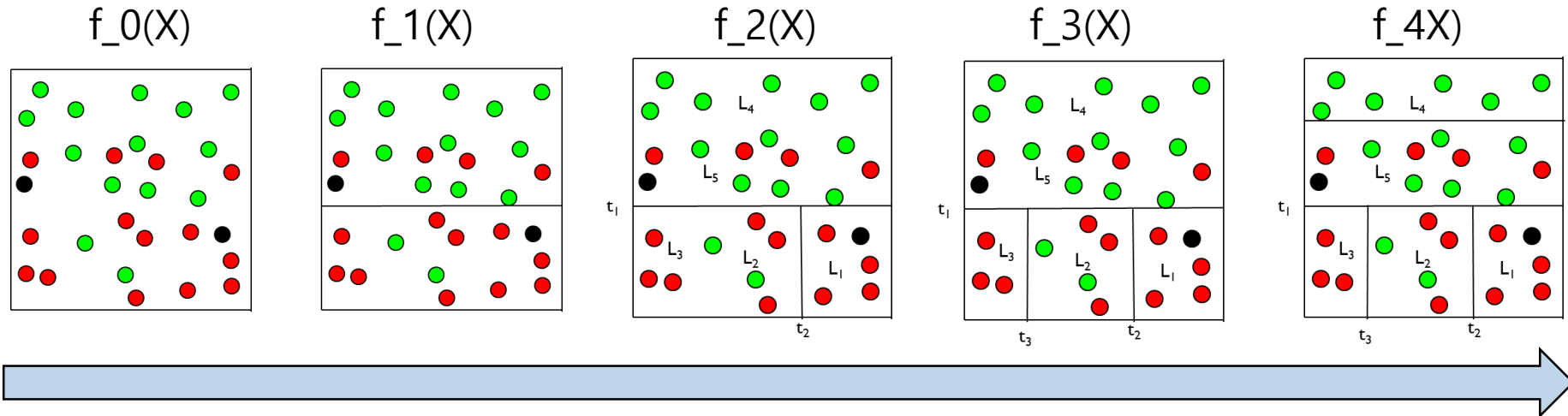
Consider tossing a coin with known, not necessarily fair, probabilities of coming up heads or tails. If heads and tails both have equal probability $\frac{1}{2}$, The entropy of the coin can be calculated as following:



$$\begin{aligned} H(X) &= - \sum_{i=1}^n p(x_i) \log_b p(x_i) \\ &= - \sum_{i=1}^2 \frac{1}{2} \log_2 \frac{1}{2} \\ &= - \sum_{i=1}^2 \frac{1}{2} \cdot (-1) = 1 \end{aligned}$$

Classification Tree – Split input space (making if-then rule)

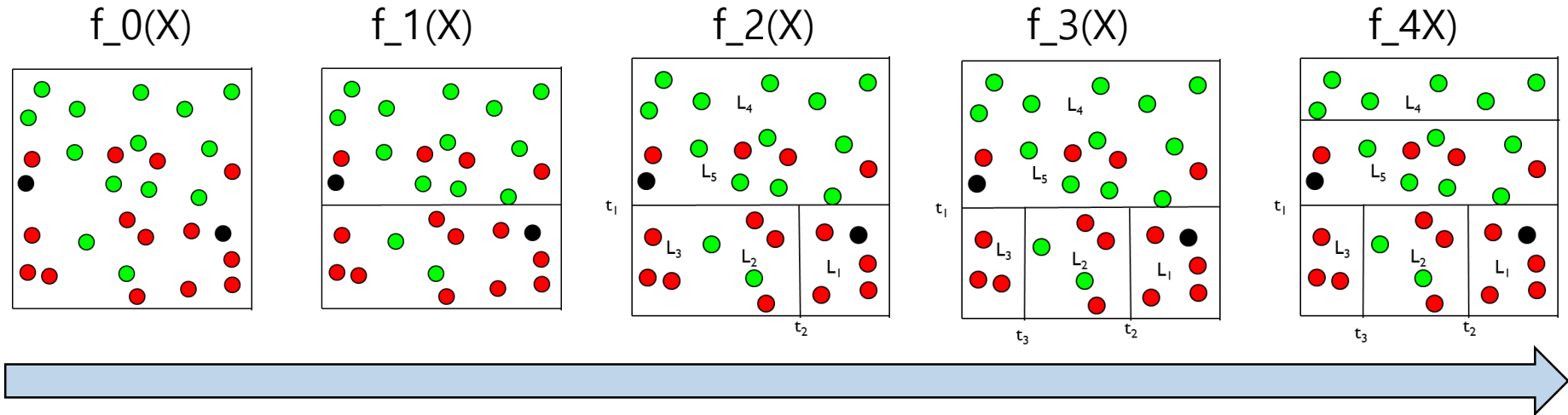
입력 변수의 영역을 두 개로 구분



$$H(X) := - \sum_{x \in \mathcal{X}} p(x) \log p(x)$$

Classification Tree – Split input space (making if-then rule)

입력 변수의 영역을 두 개로 구분



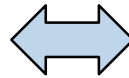
$$H(X) := - \sum_{x \in \mathcal{X}} p(x) \log p(x)$$

Minimize $_{j,s}$ Loss (j, s)

Minimize $_{j,s}$ Entropy

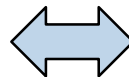
Cross Entropy Loss function

x1	x2	Y
80	28	0
60	32	0
20	91	1
25	80	1



x1	x2	$\hat{y}$
80	28	0
60	32	0
20	91	1
25	80	1

x1	x2	Y	1-Y
80	28	0	1
60	32	0	1
20	91	1	0
25	80	1	0



x1	x2	$\hat{y}$	1- $\hat{y}$
80	28	0.1	0.9
60	32	0.3	0.7
20	91	0.9	0.1
25	80	0.8	0.2

Cross Entropy Loss function

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Relation to maximum likelihood

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References

Further reading

Cross-entropy

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From Wikipedia, the free encyclopedia

In [information theory](#), the **cross-entropy** between two [probability distributions](#) p and q , over the same underlying set of events, measures the average number of [bits](#) needed to identify an event drawn from the set when the coding scheme used for the set is optimized for an estimated probability distribution q , rather than the true distribution p .

Definition [[edit](#)]

The cross-entropy of the distribution q relative to a distribution p over a given set is defined as follows:

$$H(p, q) = -\mathbb{E}_p[\log q],$$

where $\mathbb{E}_p[\cdot]$ is the [expected value](#) operator with respect to the distribution p .

The definition may be formulated using the [Kullback–Leibler divergence](#) $D_{\text{KL}}(p \parallel q)$, divergence of p from q (also known as the *relative entropy* of p with respect to q).

$$H(p, q) = H(p) + D_{\text{KL}}(p \parallel q),$$

where $H(p)$ is the [entropy](#) of p .

For [discrete](#) probability distributions p and q with the same [support](#) $\mathcal{X}$, this means

$$H(p, q) = -\sum_{x \in \mathcal{X}} p(x) \log q(x). \text{ (Eq.1)}$$

The situation for [continuous](#) distributions is analogous. We have to assume that p and q are [absolutely continuous](#) with respect to some reference [measure](#) r (usually r is a [Lebesgue measure](#) on a [Borel \$\sigma\$ -algebra](#)). Let P and Q be probability density functions of p and q with respect to r . Then

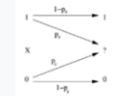
$$-\int_{\mathcal{X}} P(x) \log Q(x) \, dx = \mathbb{E}_p[-\log Q],$$

and therefore

$$H(p, q) = -\int_{\mathcal{X}} P(x) \log Q(x) \, dx. \text{ (Eq.2)}$$

NB: The notation $H(p, q)$ is also used for a different concept, the [joint entropy](#) of p and q .

Information theory



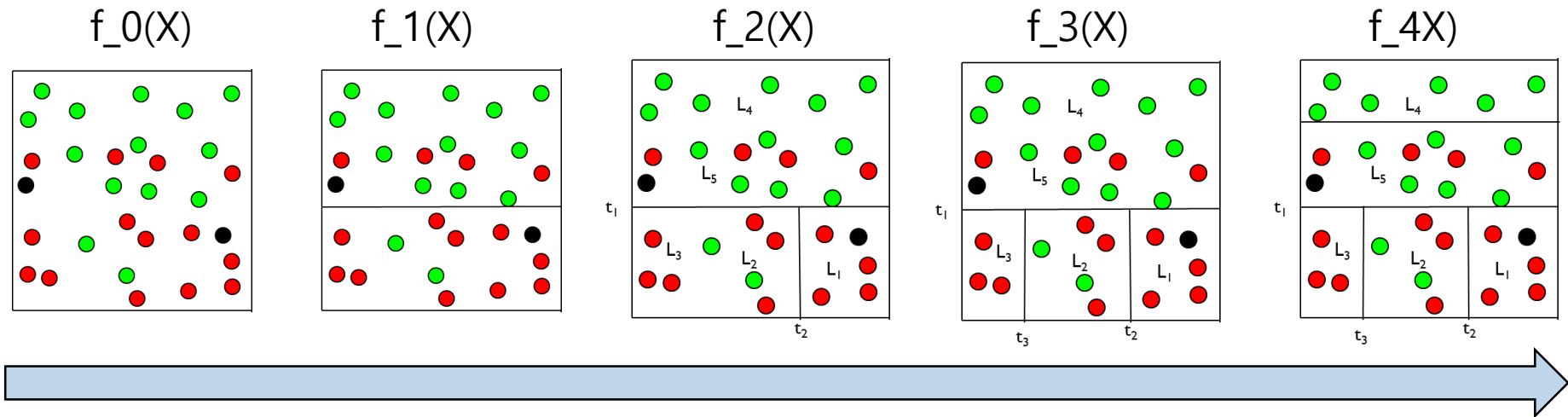
[Entropy](#) · [Differential entropy](#) · [Conditional entropy](#) · [Joint entropy](#) · [Mutual information](#) · [Directed information](#) · [Conditional mutual information](#) · [Relative entropy](#) · [Entropy rate](#) · [Limiting density of discrete points](#) · [Asymptotic equipartition property](#) · [Rate–distortion theory](#) · [Shannon's source coding theorem](#) · [Channel capacity](#) · [Noisy-channel coding theorem](#) · [Shannon–Hartley theorem](#)

V · T · E

$$\text{binary cross entropy} = -y \log \hat{y} - (1 - y) \log (1 - \hat{y})$$

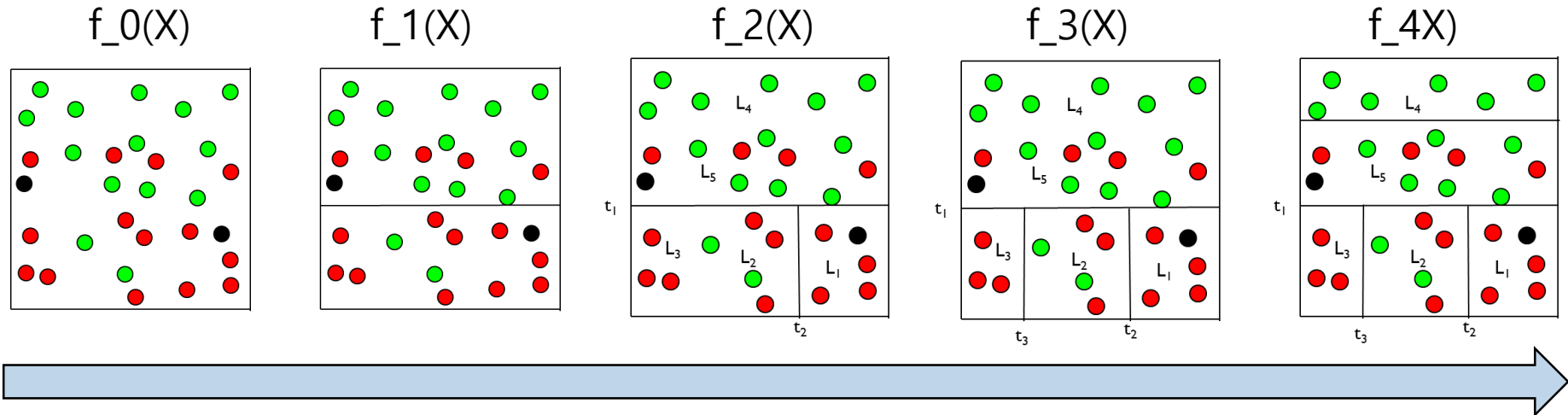
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입력 변수의 영역을 두 개로 구분



Classification Tree – Split input space (making if-then rule)

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$$H(X) := - \sum_{x \in \mathcal{X}} p(x) \log p(x)$$

Minimize _{j,s} Loss (j, s)

Minimize _{j,s} CrossEntropy